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Section : BDS-4B

Assignment:1

Operating System

This assignment is being carried out on Fedora Linux, not Ubuntu. While both are same Linux distributions, they differ slightly in 1 to 2 commands but working is same

1-Install Dependencies:

```
fkhokar@fedora:~$ sudo dnf install ncurses-devel openssl-devel
[sudo] password for fkhokar:
Last metadata expiration check: 1:13:42 ago on Fri 27 Feb 2026 07:25:27 PM PKT.
Dependencies resolved.
```

```
fkhokar@fedora:~$ sudo dnf install ncurses-devel openssl-devel
[sudo] password for fkhokar:
Last metadata expiration check: 1:13:42 ago on Fri 27 Feb 2026 07:25:27 PM PKT.
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
ncurses-devel x86_64 6.4-12.20240127.fc40 fedora 562 k
openssl-devel x86_64 1:3.2.4-1.fc40 updates 2.8 M
Installing dependencies:
ncurses-c++-libs x86_64 6.4-12.20240127.fc40 fedora 38 k
Transaction Summary
=====
Install 3 Packages

Total download size: 3.4 M
Installed size: 5.3 M
Is this ok [y/N]: y
Downloading Packages:
(1/3): ncurses-c++-libs-6.4-12.20240127.fc40.x86_ 12 kB/s | 38 kB 00:03
(2/3): ncurses-devel-6.4-12.20240127.fc40.x86_64. 118 kB/s | 562 kB 00:04
(3/3): openssl-devel-3.2.4-1.fc40.x86_64.rpm 25 kB/s | 2.8 MB 01:55
-----
Total 29 kB/s | 3.4 MB 02:00
```

```
Total 29 kB/s | 3.4 MB 02:00

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :
  Installing     : ncurses-c++-libs-6.4-12.20240127.fc40.x86_64 1/1
  Installing     : ncurses-devel-6.4-12.20240127.fc40.x86_64 1/3
  Installing     : openssl-devel-1:3.2.4-1.fc40.x86_64 2/3
  Installing     : ncurses-devel-6.4-12.20240127.fc40.x86_64 3/3
  Running scriptlet: openssl-devel-1:3.2.4-1.fc40.x86_64 3/3

Installed:
  ncurses-c++-libs-6.4-12.20240127.fc40.x86_64 ncurses-devel-6.4-12.20240127.fc40.x86_64
  openssl-devel-1:3.2.4-1.fc40.x86_64

fkhokar@fedora:~$
```

Explanation:

- **ncurses-devel**: it Provides the libraries required for the terminal interface, which is necessary for kernel configuration.
- **openssl-devel**: Provides SSL libraries, needed for kernel cryptography.

2-Directory Created for Kernel Source

```
fkhoar@fedora:~$ cd Desktop
fkhoar@fedora:~/Desktop$ sudo mkdir /kernels
[sudo] password for fkhoar:
fkhoar@fedora:~/Desktop$ cd /kernels
fkhoar@fedora:/kernels$
```

3-Download Kernel:

Download kernel using **wget** command...

```
fkhoar@fedora:~/kernels$ cd ~/kernels
wget https://cdn.kernel.org/pub/linux/kernel/v4.x/linux-4.18.9.tar.xz
Saving 'linux-4.18.9.tar.xz'
HTTP response 200 [https://cdn.kernel.org/pub/linux/kernel/v4.x/linux-4.18.9.tar.xz]
linux-4.18.9.tar.xz 100% [=====] 97.09M 920.67KB/s
[Files: 1 Bytes: 97.09M [807.70KB/s] Redirects: 0 Todo: 0 Error]
fkhoar@fedora:~/kernels$ ls
linux-4.18.9.tar.xz
```

4-Extract the kernel file:

Extract the file by using **tar -xf** command

```
fkhoar@fedora:~/kernels$ tar -xf linux-4.18.9.tar.xz
cd linux-4.18.9
fkhoar@fedora:~/kernels/linux-4.18.9$ ls
arch  COPYING  Documentation  fs      ipc      kernel  MAINTAINERS  net      scripts  tools
block CREDITS  drivers        include Kbuild   lib       Makefile     README  security  usr
certs crypto  firmware      init    Kconfig  LICENSES  mm          samples  sound    virt
fkhoar@fedora:~/kernels/linux-4.18.9$
```

5-To see config files in /boot directory:

by using command **ls and grep.**

```
fkhoar@fedora:~/kernels/linux-4.18.9$ ls /boot | grep config
config-6.12.10-100.fc40.x86_64
config-6.13.11-100.fc40.x86_64
config-6.14.5-100.fc40.x86_64
fkhoar@fedora:~/kernels/linux-4.18.9$
```

6-Copying kernel Configuration:

Here We are copying the configuration of the current running kernel to the new kernel source folder so that we don't have to reconfigure everything from scratch

```
fkhoar@fedora:~/kernels/linux-4.18.9$ sudo cp /boot/config-$(uname -r) .config
[sudo] password for fkhoar:
fkhoar@fedora:~/kernels/linux-4.18.9$
```

7-Rename the Kernel to Your Roll Number:

This will open make file

```
fkhoar@fedora:~/kernels/linux-4.18.9$ sudo gedit Makefile
```

```
# SPDX-License-Identifier: GPL-2.0
VERSION = 4
PATCHLEVEL = 18
SUBLEVEL = 9
EXTRAVERSION =
NAME = Merciless Moray
```

After Change:

```
# SPDX-License-Identifier: GPL-2.0
VERSION = 4
PATCHLEVEL = 18
SUBLEVEL = 9
EXTRAVERSION =23P-0569
NAME = Merciless Moray
```

In
above
step

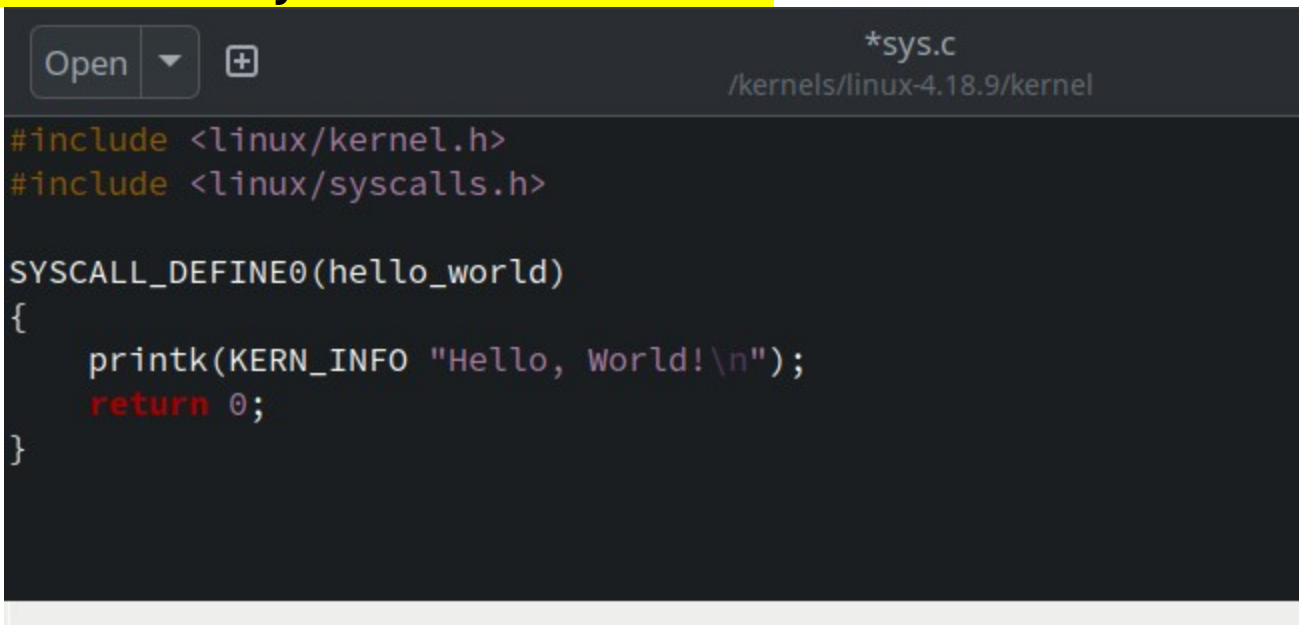
Renaming the kernel to match your roll number ensures the kernel is inamed with your roll no

8-Adding Hello world system Call:

Open sys.c file which contain souce code for system call..

```
fkhokar@fedora:~/kernels/linux-4.18.9$ sudo gedit /kernels/linux-4.18.9/kernel/sys.c
[sudo] password for fkhokar:
█
```

Function for system call "Hello world"..

A screenshot of a code editor window. The title bar shows a file named *sys.c located at /kernels/linux-4.18.9/kernel. The editor contains C code for a system call. It includes headers for linux/kernel.h and linux/syscalls.h. It defines a syscall named hello_world, which prints "Hello, World!\n" to the kernel log and returns 0. The code is as follows:

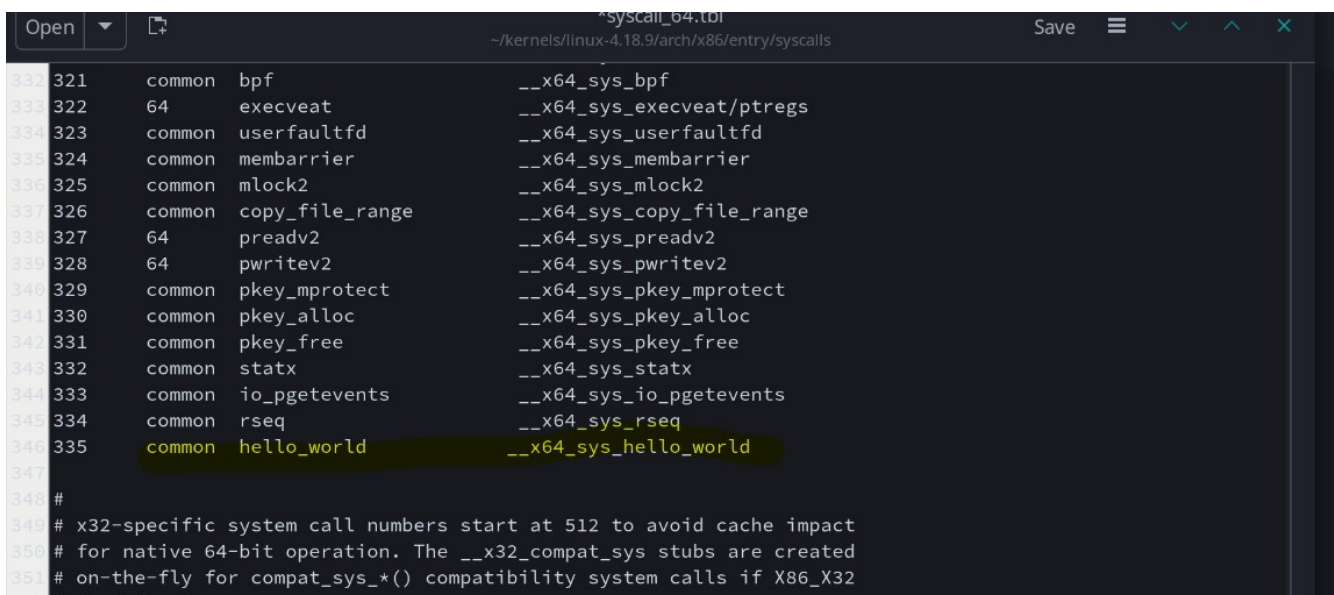
```
#include <linux/kernel.h>
#include <linux/syscalls.h>

SYSCALL_DEFINE0(hello_world)
{
    printk(KERN_INFO "Hello, World!\n");
    return 0;
}
```

Naviaget to syscall_64.tbl to Add system call in syscall table:

```
fkhoar@fedora:~/kernels/linux-4.18.9$ cd arch
fkhoar@fedora:~/kernels/linux-4.18.9/arch$ ls
alpha  arm    c6x    hexagon  Kconfig  microblaze  nds32  openrisc  powerpc  s390  sparc  unicore32  xtensa
arc    arm64  h8300  ia64     m68k     mips        nios2  parisc    riscv    sh    um    x86
fkhoar@fedora:~/kernels/linux-4.18.9/arch$ cd x86
fkhoar@fedora:~/kernels/linux-4.18.9/arch/x86$ ls
boot      entry  ia32    Kconfig    kernel  Makefile    math-emu  oprofile  power    realmode  video
configs  events  include  Kconfig.cpu  kvm     Makefile_32.cpu  mm        pci        purgatory  tools    xen
crypto    hyperv  Kbuild   Kconfig.debug  lib     Makefile.um  net       platform  ras      um
fkhoar@fedora:~/kernels/linux-4.18.9/arch/x86$ cd entry
fkhoar@fedora:~/kernels/linux-4.18.9/arch/x86/entry$ ls
calling.h  entry_32.S      entry_64.S  syscall_32.c  syscalls    thunk_64.S  vsyscall
common.c  entry_64_compat.S  Makefile    syscall_64.c  thunk_32.S  vdso
fkhoar@fedora:~/kernels/linux-4.18.9/arch/x86/entry$ cd syscalls
fkhoar@fedora:~/kernels/linux-4.18.9/arch/x86/entry/syscalls$ ls
Makefile  syscall_32.tbl  syscall_64.tbl  syscallhdr.sh  syscalltbl.sh
fkhoar@fedora:~/kernels/linux-4.18.9/arch/x86/entry/syscalls$ cd syscall_64.tbl
bash: cd: syscall_64.tbl: Not a directory
fkhoar@fedora:~/kernels/linux-4.18.9/arch/x86/entry/syscalls$ gedit syscall_64.tbl
```

this adds a new system call that prints "Hello, World" in the kernel logs using printk.



```
*syscall_64.tbl
~/kernels/linux-4.18.9/arch/x86/entry/syscalls

332 321    common  bpf                __x64_sys_bpf
333 322    64      execveat           __x64_sys_execveat/ptregs
334 323    common  userfaultfd        __x64_sys_userfaultfd
335 324    common  membarrier         __x64_sys_membarrier
336 325    common  mlock2             __x64_sys_mlock2
337 326    common  copy_file_range    __x64_sys_copy_file_range
338 327    64      preadv2            __x64_sys_preadv2
339 328    64      pwritev2           __x64_sys_pwritev2
340 329    common  pkey_mprotect       __x64_sys_pkey_mprotect
341 330    common  pkey_alloc          __x64_sys_pkey_alloc
342 331    common  pkey_free           __x64_sys_pkey_free
343 332    common  statx              __x64_sys_statx
344 333    common  io_pgetevents       __x64_sys_io_pgetevents
345 334    common  rseq               __x64_sys_rseq
346 335    common  hello_world         __x64_sys_hello_world
347
348 #
349 # x32-specific system call numbers start at 512 to avoid cache impact
350 # for native 64-bit operation. The __x32_compat_sys stubs are created
351 # on-the-fly for compat_sys_*() compatibility system calls if X86_X32
352 # is defined.
```

9-Configuration of Kernel:

here we copy running configuration kernel parameter to our kernel

```
fkhokar@fedora:~/kernels/linux-4.18.9$ yes '' | make oldconfig -j4
scripts/kconfig/conf -s --oldconfig Kconfig
fkhokar@fedora:~/kernels/linux-4.18.9$ yes n | make oldconfig -j4
scripts/kconfig/conf -s --oldconfig Kconfig
fkhokar@fedora:~/kernels/linux-4.18.9$ yes n | make oldconfig -j4
scripts/kconfig/conf -s --oldconfig Kconfig
fkhokar@fedora:~/kernels/linux-4.18.9$ yes n | make oldconfig -j4
scripts/kconfig/conf -s --oldconfig Kconfig
fkhokar@fedora:~/kernels/linux-4.18.9$ yes '' | make oldconfig -j4
scripts/kconfig/conf -s --oldconfig Kconfig
fkhokar@fedora:~/kernels/linux-4.18.9$ yes n | make oldconfig -j4
scripts/kconfig/conf -s --oldconfig Kconfig
fkhokar@fedora:~/kernels/linux-4.18.9$ make oldconfig -j4
scripts/kconfig/conf -s --oldconfig Kconfig
fkhokar@fedora:~/kernels/linux-4.18.9$
```

10-cleaning temporary files before building kernel...

```
fkhokar@fedora:~/kernels/linux-4.18.9$ make clean -j4
fkhokar@fedora:~/kernels/linux-4.18.9$
```

After configuring the kernel, I faced issues when running `make bzImage -j4`, including errors related to kernel headers and SELinux. instead of my efforts, I could not resolve all the errors in time for the assignment submission.